University of Baghdad
College of Medicine2025-2026

Title: Respiratory Failure Grade: 4th grade

Module: Respiratory Medicine Module

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Date:

Definition

Abnormal blood gases due to diseases of the respiratory system or its control (i.e. : respiratory centre) when the patient is breathing at the sea level & at rest.

Physiology



For proper respiration, you need to have 3 important functions :

1- Proper ventilation.

2- Proper perfusion.

3- Proper exchange of gases.

Classifications

Clinically, respiratory failure is classified into 2 types depending on the presence or absence of Hypercapnoea.

Type I : Hypoxia is associated with Normo or Hypocapnoea.

Type II : Hypoxia is associated with Hypercapnoea.

Type I Respiratory Failure

The main pathogenetic factor in the production of type I RF is Ventilation-Perfusion Mismatching.

Causes

1- Restrictive defects of ventilation

* Infiltrative lung diseases.
* Pneumonias.
* Cardiac pulmonary edema.
* ARDS.
* Sudden pneumothorax.
* Sudden pleural effusion.

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2- Obstructive defects of ventilation

* Acute asthmatic attack.
* Acute bronchiolitis.

3- Vascular causes

* Pulmonary thromboembolism.

* Pulmonary vasculitis.

Clinical Features

* Sings & symptoms of the underlying cause.

* Sings & symptoms of hypoxia as : SOB, restlessness, agitation, confusion, & central cyanosis.

Here, the patient has progressive SOB associated with tachypnoea (shallow & rapid breathing), & the central cyanosis is obvious on minimal exertion that can be corrected by O2 therapy for more than 30% concentration.

Diagnosis

* The Dx is mostly clinical.
* Dx of the underlying cause.
* Lab. Investigations.
* Radiological investigations.
* & this is confirmed by arterial blood gases analysis.

Treatment

1- Rx of the underlying cause.

2- General Rx :

* Rest in bed.
* O2 therapy (Liberal).

Type II Respiratory Failure

The main pathogenetic factor in the production of type II RF is Hypoventilation.

Causes

1- COPD.

2- Hypodynamic causes :

* Respiratory centre depression (Trauma, Tumors, Vascular accident, Anesthesia).
* Nerves' affection (Intercostal & Phrenic n.) as in : Gullian-Barre syndrome, Poliomyelitis
* Muscles, as in : Myopathies & Myasthenia Gravis.
* Thoracic cage, as in : Kyphoscoilosis & flail chest.

3 -ىEnd stage of type I respiratory failure

Causes

1- COPD.

2- Hypodynamic causes :

* Respiratory centre depression (Trauma, Tumors, Vascular accident, Anesthesia).
* Nerves' affection (Intercostal & Phrenic n.) as in : Gullian-Barre syndrome, Poliomyelitis
* Muscles, as in : Myopathies & Myasthenia Gravis.
* Thoracic cage, as in : Kyphoscoilosis & flail chest.

3 -ى End stage of type I respiratory failure

Clinical Features

1- The clinical features of the underlying cause.

2- The clinical features of Hypoxia.

3- The clinical features of Hypercapnoea :

* Headache.
* Palpitation.
* Tremor.
* Drowsiness.
* Confusion & coma.

Diagnosis

1- Clinically.

2- Investigations :

* Lab. investigations.
* Radiological investigations.
* Arterial blood gases analysis.

Treatment

1- Rx of the cause.

2- General measures that improve alveolar ventilation & it means improvement of the hypoxia & hypercapnoea.

These General measures include :

* O_2 Therapy.
* Airway clearance.
* Assisted ventilation.

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